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Palaeomagnetic Studies in the British Caledonides—I Igneous Rocks of the Builth Wells–Llandridnod Wells Ordovician Inlier, Radnorshire, Wales

J. D. A Piper and J. C. Briden

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Summary

Comprehensive sampling of intrusive and extrusive volcanic rocks in Radnorshire, Wales (52·1° N, 3·3° W) leads to two new estimates of Ordovician palaeomagnetic field directions. The mean direction of magnetization for the Builth Volcanic Series of Upper Llanvirn age is $\bar{D} = 177°$, $\bar{I} = +39°$ ($\alpha_{95} = 8·9°$) based on 15 sites. In minor basic intrusions of early Ashgillian age two mutually antiparallel groups of three sites yield a mean direction irrespective of polarity of $\bar{D} = 174°$, $\bar{I} = +55°$ ($\alpha_{95} = 11·2°$) which is regarded as a best estimate although results from five other statistically significant sites have to be explained as transitional during polarity change or disregarded as anomalous. The polarity transition recorded in the intrusives is likely to be $R \rightarrow N$ as inferred from the probable sequence of intrusion. All results are presented after magnetic cleaning and correction for the regional westward tilt of the inlier. Supporting thermal demagnetization data are presented. Small scale folding in the vicinity of the Welfield intrusion is shown to have originated as part of the intrusion process, and the data are compatible with the postulate that the Welfield intrusion is laccolithic. Pole positions from the volcanics and intrusives are respectively 359° E, 16° S ($d\psi = 6°$, $d\chi = 11°$) and 2° E, 2° S ($d\psi = 11°$, $d\chi = 16°$). The results are significantly different at the 95 per cent confidence level but discussion of this difference is deferred to Paper VI.

1. Introduction

This paper presents the results of a palaeomagnetic survey of volcanic rocks of the Ordovician inlier between Builth Wells and Llandridnod Wells in Radnorshire referred to in this paper as the 'Builth inlier'. The area was an active volcanic centre during part of the Ordovician period, and the rocks accumulated during this interval were intruded by dolerites and tilted before the deposition of the Silurian (Wenlock) rocks which now surround the inlier. The volcanic rocks are well exposed, unmetamorphosed, structurally simple, and accurately dated by the graptolite fauna of the included shales.

Geological studies of the inlier have been summarized by Jones & Pugh who mapped the complete inlier on the scale of 25 inches to 1 mile; a preliminary account

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appeared in 1941 and was followed in 1946, 1948 (a, b) and 1949 by papers concerned with particular aspects of the geology. Their detailed work remains the principal source of information on the geology of the inlier though several specialized studies (e.g. Nicholls 1958) have appeared since. Nesbitt (1967) has reported palaeomagnetic results from nine samples of lavas from the inlier.

2. Geology

During the Ordovician, sediment deposition in central Wales was characterized by black shale and mudstones with frequent interruptions marked by disconformities. The Builth–Llandridnod Wells area was a zone of active volcanism during Upper Llanvirn times and the Builth Volcanic Series (Jones & Pugh 1949) spans the graptolite zone of *Didymograptus murchisoni*. This series is best developed in the southern part of the inlier, and thins rapidly northwards, being represented only by a tuff horizon in the vicinity of Llandridnod Wells. In the southern part of the inlier, sampled in the present study, it comprises about 400 m of tuffs and agglomerates overlain by approximately 200 m of spilite and keratophyre lava flows. The whole volcanic suite is highly sodic and was extensively metasomatized during formation to a greenschist facies mineral assemblage. It was laid down underwater and the pyroclastic sequence includes shale horizons showing much evidence of reworking; the spilites and keratophyres occur as numerous thin flow units, some showing pillow structures and some separated by shale horizons. The volcanic succession was extensively intruded by small bodies of andesite, uplifted and partially eroded shortly after formation. The Newmead Series which lies unconformably on the volcanic complex is a littoral facies developed in the upper part of the zone of *D. murchisoni* (Jones & Pugh 1949). This is in turn overlain by black shales and mudstones of Llandeilo age.

Intrusion of the Llanvirn–Llandeilo rocks by dolerites took place in Upper Bala (early Ashgillian) time (Jones & Pugh 1948b). Like the lavas, these minor intrusions were extensively metasomatized at the time of their formation. Following the intrusion of the dolerites the entire sequence was tilted to acquire an inclination of 30–40° to the west and subjected to dextral tear faulting before the deposition of the earliest Silurian rocks in Upper Llandovery time.

Samples were collected at 15 sites in the Builth Volcanic Series and 12 sites in the Ashgillian minor intrusions in the form of short cores using a drill, or as hand samples from which cores were drilled in the laboratory. Wherever possible orientations were made with a sun compass and when this was not feasible they were oriented to magnetic co-ordinates. The Natural Remanent Magnetizations (NRMs) were measured using a PAR SM-2D magnetometer and an alternating field demagnetizer similar to that described by McElhinny (1966). One sample from each site was partially thermally demagnetized in the Nuffield Laboratory of Newcastle University. Magnetic susceptibilities were measured with a bridge of the kind described by Collinson & Molyneux (1967).

3. Palaeomagnetic results

3.1 *Total NRM intensities and susceptibilities*

Of the 15 sites in the Builth Volcanic Series ten (sites 6–15) are in spilite and keratophyre lavas and five (sites 1–5) are in tuffs. These latter five sites were chosen to avoid the visibly reworked parts of the succession. There is no significant difference in the magnetic properties of the two groups. The lavas and tuffs have intensities of magnetization varying between $0{\cdot}8\times10^{-4}$ and $0{\cdot}4\times10^{-6}$ Gauss with a logarithmic mean intensity of $1{\cdot}55\times10^{-6}$ Gauss (80 samples). The dolerite intrusions tend to be rather weaker (Fig. 1(c)) with intensities varying between

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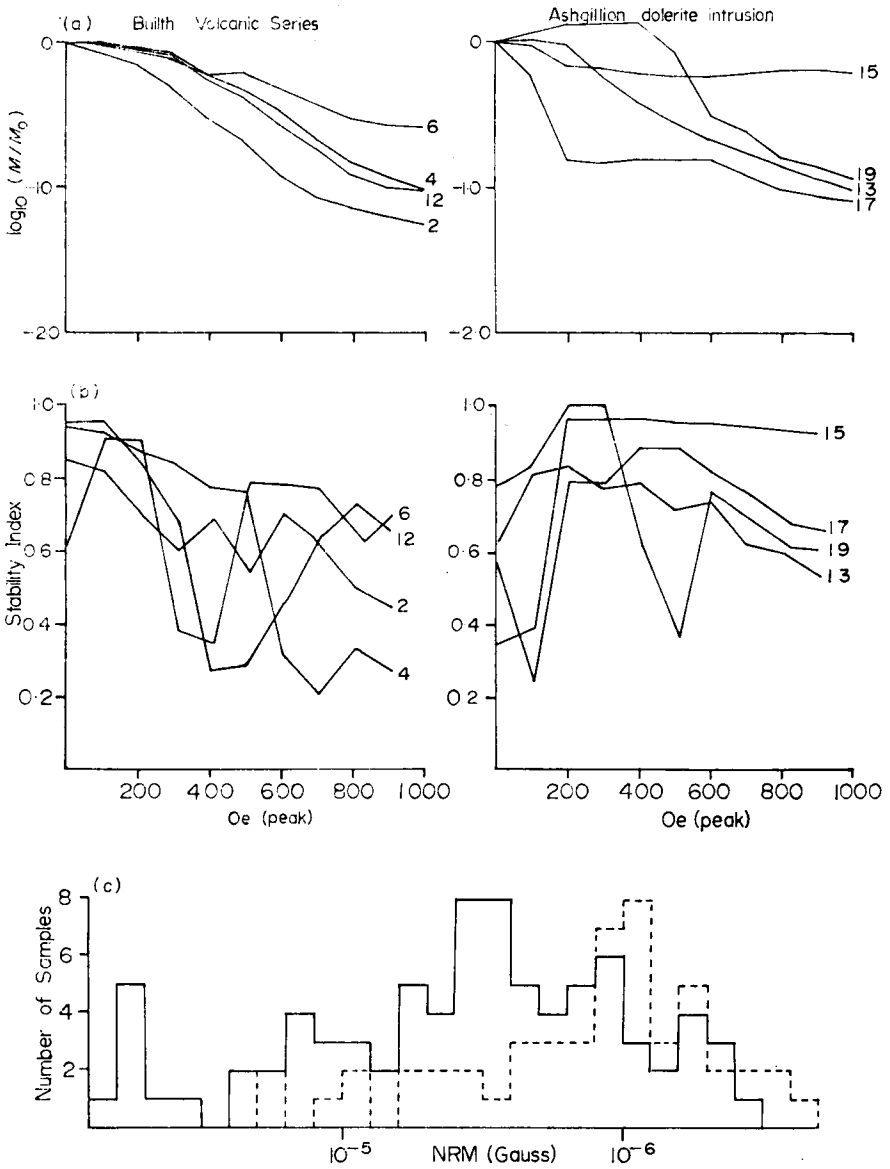


FIG. 1. (*a*) AF demagnetization curves for four samples each from the Builth Volcanic Series and from the Ashgillian dolerite intrusions. (*b*) The variation in Briden's (1972) stability index with applied demagnetizing field for the same samples. (*c*) Frequency histogram of magnetizations of 80 samples of the Builth Volcanic Series (solid line) and 51 samples of the Ashgillian dolerite intrusions (dashed line).

$0{\cdot}2\times10^{-4}$ to $0{\cdot}3\times10^{-6}$ Gauss with logarithmic mean of $1{\cdot}13\times10^{-6}$ Gauss (51 samples). The observation that extrusives are more strongly magnetized than comparable intrusives has been noted in Precambrian rocks by Piper (1972) and previously in Tertiary rocks by Ade-Hall *et al.* (1970) who used it as an argument in favour of a primary origin of remanence. The susceptibilities of the tuffs, lavas, and intrusions are comparable; the tuffs and lavas have a logarithmic mean susceptibility of $2{\cdot}25\times10^{-5}$ Gauss/Oersted (78 samples) and the intrusions have a

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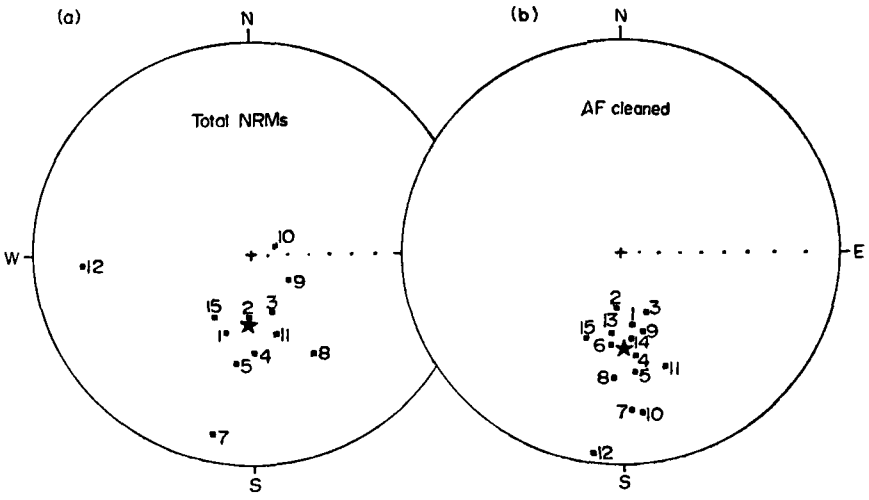


FIG. 2. Dip-corrected directions of magnetization of sites in the Builth Volcanic Series (*a*) before and (*b*) after AF cleaning. The star is the mean direction of magnetization. Throughout this paper solid symbols denote downwards, and open symbols denote upwards.

logarithmic mean susceptibility of $2{\cdot}45 \times 10^{-5}$ Gauss/Oersted (71 samples). Nesbitt (1967) has commented on the weak magnetic properties of the Builth Volcanic Series and has found comparable values to those observed here. The intensities and susceptibilities suggest that the weak properties are a function of the mineralogy, and are consistent with the petrological observations of Agrell (in Jones & Pugh 1946) and Nicholls (1958) that the bulk of the opaques are non-magnetic ilmenite and pyrite.

3.2 *Total NRM directions*

Twelve of the 15 sites in the Builth Volcanic Series show groupings of total NRM directions (Table 1, Fig. 2(a)) which are significant at the 95 per cent confidence level (Watson 1956; Irving 1964). All but sites 10 and 12 have a southerly declination and positive inclination, and all sites combine to give a direction $D = 184°$, $I = +53°$, $\alpha_{95} = 19°$ (Fisher 1953) after correction for a mean regional dip of 40° to the west. Nine of the 12 sites in the minor intrusions of Ashgillian age give groupings which are significant with 95 per cent confidence (Table 2 and Fig. 3(a)). Site 17 is anomalous and sites 16 and 21 are magnetized nearly antiparallel to the remainder. Excluding site 17 and reversing sites 16 and 21 gives a combined direction of $D = 142°$, $I = +83°$, $\alpha_{95} = 46°$.

3.3 *Alternating field (AF) demagnetization results*

Progressive AF demagnetization has been carried out on one or two specimens from each site. Most specimens from the Builth Volcanic Series show a gradual and approximately exponential decrease in their moments accompanied by little change in the direction of NRM (Figs 1(a) and 4(a)). This behaviour is reflected in the specimens showing highest stability indices (SI) (Briden 1972) in fields less than about 300 Oe, above which the value of the index tends to fall (Fig. 1(b)). The directional stability of the NRM is however, preserved to fields of 600–1000 Oe. In higher alternating fields several specimens (e.g. pilots of sites 1 and 13) tend to acquire new magnetizations due to limitations of the demagnetizing apparatus.

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Table 1

Builth Volcanic Series, site statistics after dip correction

		Total NRM						After magnetic cleaning							
Site	No. of specimens N	R	D	I	k	α_{95}	Peak Alt. field (Oe)	No. of specimens N	R	D	I	k	α_{95}	Virtual Geomagnetic North Pole Long.	Lat.
1	7	5·61	199	+48	4·3	32·1	300	6	5·92	171	+53	60·8	8·6	4° E	4° S
2	3	2·96	182	+59	54·1	16·9	200	3	2·96	186	+60	48·0	18·0	352° E	3° N
3	3	2·97	159	+59	69·5	14·9	200	3	2·97	157	+57	76·3	14·2	14° E	2° N
4	6	5·91	179	+40	55·3	9·1	200	6	5·95	172	+39	101·8	6·8	4° E	16° S
5	5	4·94	188	+36	66·1	9·5	300	5	4·98	174	+32	176·8	5·8	3° E	20° S
6	5	3·20	—	—	—	—	200	5	4·56	187	+43	9·0	26·9	35° E	13° S
7	6	5·88	193	+8	41·7	10·5	300	6	5·68	176	+18	15·7	17·5	1° E	29° S
8	6	5·70	149	+34	16·7	16·9	250	6	5·84	184	+30	30·4	12·3	353° E	22° S
9	5	4·29	124	+67	5·6	35·3	250	4	3·89	166	+50	27·2	18·0	9° E	6° S
10	5	4·85	64	+76	26·5	15·1	150	4	3·92	173	+17	35·8	15·6	5° E	29° S
11	5	4·88	163	+48	33·8	13·4	200	5	4·72	160	+31	14·4	20·9	17° E	19° S
12	7	5·73	268	+16	4·7	31·0	200	6	5·20	188	+3	6·2	29·1	347° E	36° S
13	3	2·51	—	—	—	—	300	3	2·92	187	+49	26·6	24·4	351° E	8° S
14	4	2·84	—	—	—	—	200	3	2·97	174	+47	63·2	15·6	3° E	10° S
15	3	2·82	210	+53	11·2	38·6	300	3	2·88	203	+44	17·4	30·5	336° E	10° S
							Total NRM 12 sites		10·18	184	+53	6·1	19·2	353° E	4° S
														$(d\psi, d\chi) = (18·2, 26·5)$	
							AF cleaned 15 sites		14·28	177	+39	19·4	8·9	359° E	16° S
														$(d\psi, d\chi) = (6·3, 10·6)$	

Table 2

Ashgillian minor intrusives, site statistics after correction for region dip

		Total NRM						After magnetic cleaning							
Site	No. of specimens N	R	D	I	k	α_{95}	Peak Alt. field (Oe)	No. of specimens N	R	D	I	k	α_{95}	Virtual Geomagnetic North Pole Long.	Lat.
16	3	2·98	342	−59	113·2	11·6	800	3	2·98	2	−56	107·8	11·9	175° E	1° N
17	3	2·97	344	−11	59·0	16·2	800	3	2·99	23	−51	758·3	4·5	157° E	3° N
18	4	3·05	—	—	—	—	500	6	4·49	354	−13	3·3	43·6	185° E	31° N
19	3	2·95	16	+51	38·9	20·0	500	3	2·94	338	+69	39·0	20·0	274° E	77° N
20	4	2·91	—	—	—	—	800	4	2·84	—	—	—	—	—	—
21	4	3·57	331	−41	7·1	37·3	300	4	3·98	351	−45	170·8	7·1	184° E	11° N
22	7	5·87	306	+34	5·3	28·9	200	5	4·86	311	+15	28·5	14·6	237° E	30° N
23	2	1·99	75	+79	659·0	9·7	800	2	1·97	55	+74	35·4	43·3	50° E	59° N
24	5	4·92	66	+71	52·1	10·7	800	5	4·64	68	+57	11·4	23·7	76° E	31° N
25	4	2·65	—	—	—	—	400	4	3·86	166	+69	22·0	20·0	5° E	16° N
26	5	4·89	154	+62	37·9	12·6	200	5	4·84	157	+58	25·1	15·6	15° E	3° N
27	3	2·91	164	+51	23·2	26·2	200	3	2·95	162	+46	4·1	19·4	13° E	9° S
Total NRM (all significant sites) 9 sites									5·4	142	+83	2·1	46·2	8° E	40° S
														$(d\psi, d\chi) = (88, 90)$	
AF cleaned (all significant sites except 18, 22) 9 sites									8·0	160	+70	8·4	18·9	9° E	17° N
														$(d\psi, d\chi) = (28, 32)$	
AF cleaned (also excluding 19, 23, 24) 6 sites									5·9	174	+55	36·7	11·2	2° E	2° S
														$(d\psi, d\chi) = (11, 16)$	

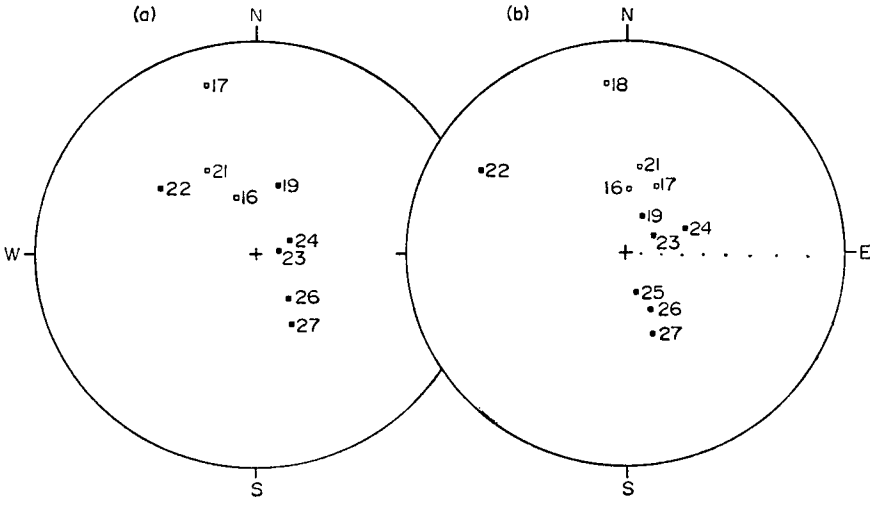


FIG. 3. Dip-corrected site-mean directions of magnetization of the Ashgillian dolerite intrusions (*a*) before, and (*b*) after AF cleaning.

The dolerite intrusions show less systematic behaviour on AF demagnetization. In several specimens (e.g. pilot of site 19, Figs 1(a) and 4(b)) the remanence direction moves away from the present Earth's field while magnetic moment increases in fields up to a few hundred Oersteds. This is consistent with selective removal of a geologically recent viscous component. Other specimens (e.g. pilot of site 17, Fig. 1(a)) show an initial large fall in moment accompanied by large directional changes (Fig. 4(a)). This is followed by a more gradual, exponential decrease which corresponds to a steady remanence direction. This behaviour is probably due to Isothermal Remanent Magnetization (IRM) superimposed on the stable remanence. The selective removal of secondary components of either Viscous

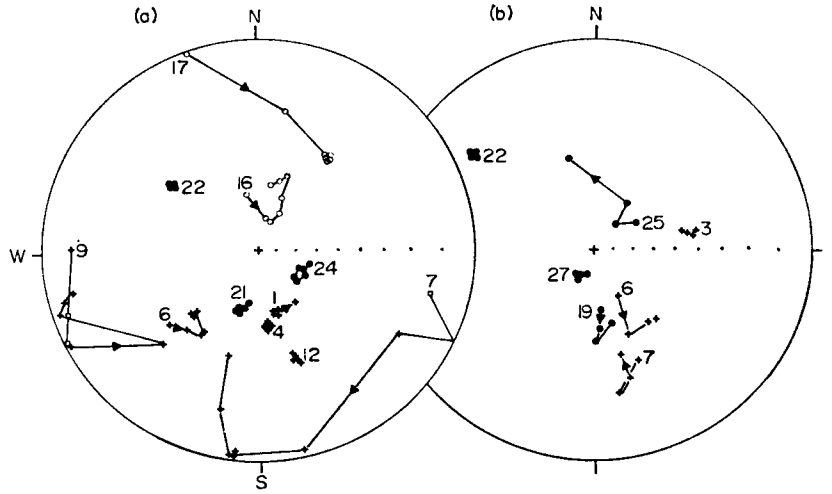


FIG. 4. (*a*) Progressive AF demagnetization of samples from the Builth Volcanic Series (solid crosses and open squares) and the Ashgillian dolerite intrusions (circles), (*b*) Progressive thermal demagnetization of samples from both collections in steps of 200, 320 and 450 °C.

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Remanent Magnetization (VRM) or IRM origin is also reflected in the increase of SI with progressive demagnetization (Fig. 1(b)). optimum values being attained in the range 200–800 Oe (peak).

3.4 *Thermal demagnetization*

One specimen from each site has been partially thermally demagnetized at steps steps of 200, 320 and 450 °C. Intensities of magnetization in most cases fall only gradually with this treatment and directions are in general accord with those observed after AF cleaning (Fig. 4(b)).

3.5 *Mean directions and palaeomagnetic poles*

Specimens from all sites in the Builth Volcanic Series have been cleaned in peak fields of up to 300 Oe. The three sites which showed no statistically significant grouping of total NRMs (6, 13, 14) become significant after cleaning, and in no case does within-site grouping deteriorate markedly upon cleaning (McElhinny (1964; Cox (1969)). All the treated mean directions combine to give an overall mean $D = 177°$, $I = +39°$ with $\alpha_{95} = 9°$ corresponding to a palaeomagnetic pole at 359° E, 16° S.

The dolerite intrusions were demagnetized in fields of 200–800 Oe (peak) after which all sites except 20 are significant (Fig. 3(b)). Several sites, notably 17 and 21, show a dramatic increase in precision after cleaning and in only one case (24) does

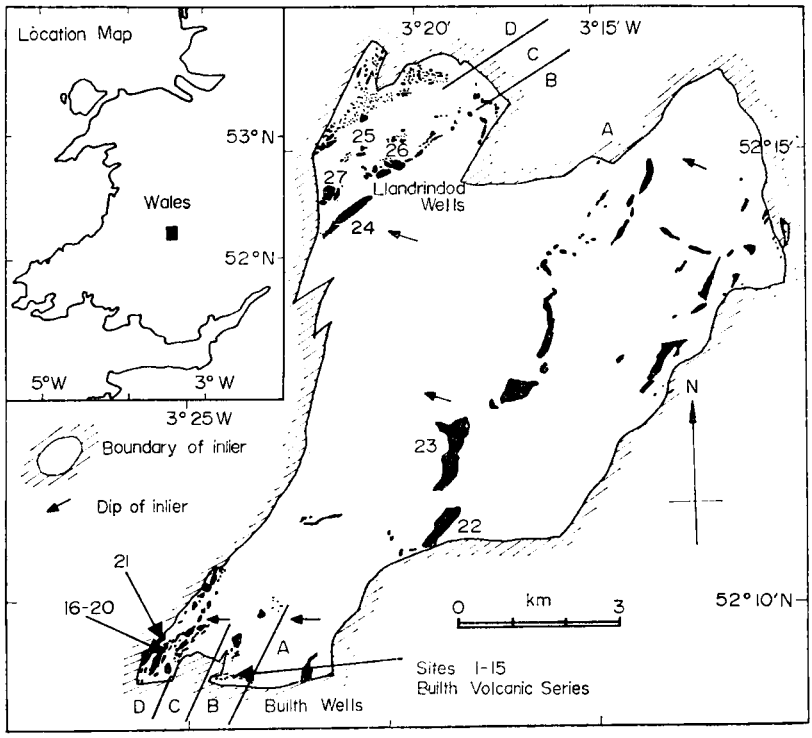


FIG. 5. The distribution of the Ashgillian dolerite intrusions (in black) and palaeomagnetic sites within the Builth Wells-Llandridnod Wells Ordovician inlier. The letters A, B, C and D refer to successively higher graptolite zones *Didymograptus bifidus*, Lower *Glyptograptus teretiusculus*, Upper *G. teretiusculus*, and *Nemagraptus gracilis* respectively. The location of the sampling sites in the Builth Volcanic Series is indicated, although the outcrop of the volcanics cannot be satisfactorily illustrated on this map.

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precision deteriorate notably (McElhinny 1964; Cox 1969). Two of the sites (18, 22) have low inclinations and their directions are widely removed from the remainder of the collection. If these are excluded there remain three groups, each of three sites. The group comprising sites 16, 17 and 21 is closely antiparallel to that of sites 25, 26, 27. Accepting this reversal the best estimate of the direction of magnetization of these minor intrusions, is $D = 174°$, $I = +55°$ with $\alpha_{95} = 11°$.

There remains the problem of explaining the anomalous group of sites 19, 23 and 24. They are geographically widely separated (Fig. 5) and their stability is comparable to the remainder. Their dip-corrected remanence direction does not accord with the group of six sites in these intrusions, nor with that predicted by other studies in the British Ordovician (Papers II and III). Moreover their *in situ* stable

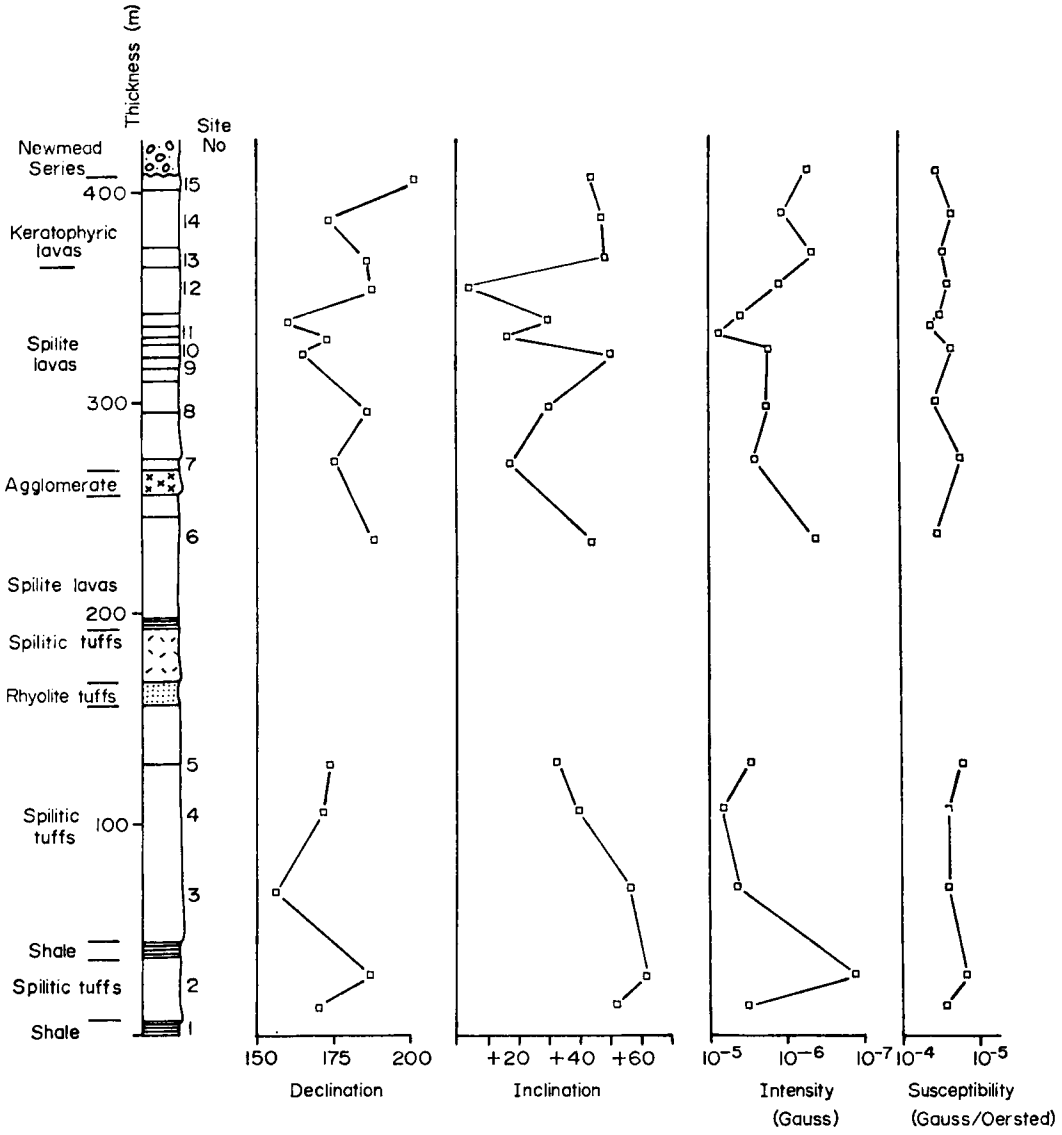


FIG. 6. The variation in total remanence and susceptibility through a measured sequence in the Builth Volcanic Series.

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remanence does not match the present geomagnetic field nor the inferred local palaeomagnetic field at any time since the tilting of these rocks. We are forced to regard them as an intermediate group, comparable to those found in Lower Devonian lavas in Scotland (Paper IV); some evidence for this interpretation is given in Section 4.2 where it is argued that site 19 may have been magnetized during a geomagnetic polarity transition.

4. Discussion

4·1 *Magnetic stratigraphy*

The distribution of the 15 sites through a measured section of the Builth Volcanic Series is shown in Fig. 6. All the members defined by Jones & Pugh (1949) were sampled except the basal Fine Ash and Shales, and the Red Agglomerate and Ash. The declination throughout the succession is southerly and points to a virtual geomagnetic north pole falling on the *south* polar wander path for Britain (Paper VI): hence the sequence is regarded as being of reversed polarity. The only obvious correlations between magnetic properties in Fig. 6 are at site 2 which of all sites shows maximum inclination and minimum intensity, and site 10 which shows minimum inclination, maximum intensity and susceptibility (though susceptibility variation is small).

4.2 *Reversal of magnetization within the Ashgillian intrusions*

The distribution of the dolerite intrusions within the inlier has been examined in detail by Jones & Pugh (1949). They have noted that the intrusions become progressively smaller and more subdivided in the higher graptolite zones and ascribe these bodies to one phase of intrusion into as yet unlithified and horizontal Llanvirn and Llandeilo sediments. On this interpretation the deeper level intrusions, now exposed in the south-east part of the inlier (sites 22 and 23), were injected before the medium (sites 24–27) and then the high level bodies (sites 16–21, Fig. 5). The palaeomagnetic results are consistent with this picture in so far as the middle and low level intrusions are all magnetized with steep positive inclinations. The lowest site (19) in the high level group is similarly magnetized, the next lowest (18) is magnetized with an anomalous direction, and the highest level group (16, 17 and 21) are magnetized with a negative inclination and are apparently reversed with respect to the lowest intrusions. Site 18 is also exceptionally weakly magnetized with a total NRM intensity of $0{\cdot}5\times10^{-7}$ Gauss, further suggesting that it might have been magnetized during a reversal. Using the polar wander curve (Paper VI, Fig. 1) as reference, the reversal is in the sense $R \rightarrow N$.

This reversal takes place within the complex intrusion at Welfield, 2 km from Builth Wells (sites 16–21); all the sites except 21 are located in the section exposed by the River Wye (Fig. 7(a)). The problem here is whether the attitude of each component body at the time of intrusion is to be reconstructed by restoring the regional dip of the inlier to the horizontal (as was the case with the Builth Volcanic Series), or whether the more variable local folding of the sediments immediately intruded by the dolerites should be taken into account. Fig. 7(b) shows that the grouping of sites 16, 17 and 21 is much closer after correction of the regional dip to the horizontal (denoted by circles) than it is if the palaeohorizontal is identified as the bedding plane of the nearest sediments (results denoted by squares in Fig. 7(b)). This implies that these sediments were folded before the rocks acquired their *magnetism*. Since Jones & Pugh (1946, 1948b) argue that the country rocks were horizontal and unfolded at the time of *intrusion* the palaeomagnetic results lead us to conclude that the process of intrusion accompanied the deformation of the intruded sediments. In other words, our results support the suggestion of Jones & Pugh (1946) that the Welfield intrusion was originally of laccolithic form.

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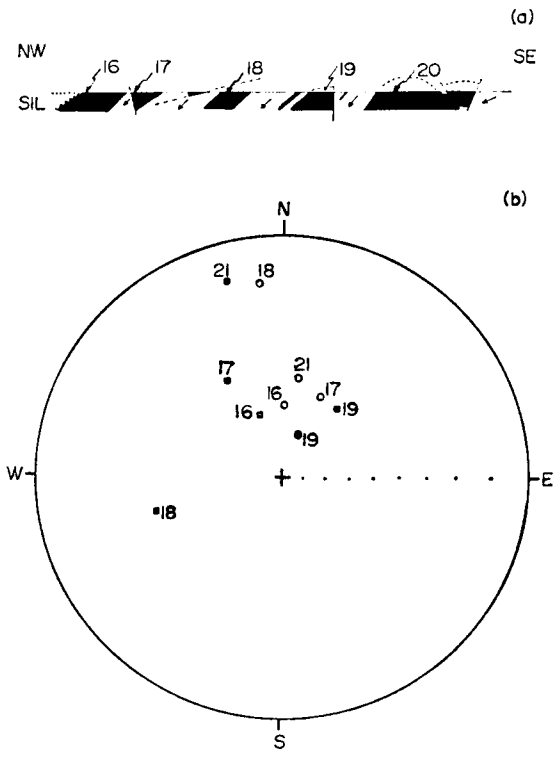


FIG. 7. (*a*) Location of sampling sites in the Welfield intrusion near Builth Wells, based on Jones & Pugh (1946, Fig. 5). The exposures are along the course of the River Wye. The dotted line is the extrapolated top of the Welfield complex. The section is NW–SE, i.e. aligned with the regional dip illustrated by the pre-Silurian unconformity at the NW end; but the Ordovician sediments also have a dip component oblique to the line of section, and their dip actually varies much more than the diagram shows. Removal of the regional tilt shows that the pre-Silurian *strike* was more or less in the plane of this section. (*b*) Directions of magnetization for the significant sites corrected for the regional dip of the inlier (circles) and additionally for the dip of the adjacent sediments (squares).

5. The pole position in Ordovician times

There are several lines of evidence to suggest that the mean NRM directions of the two suites of rocks studied are indicative of the local Ordovician geomagnetic field. Principally the remanences are pre-tilting because their *in situ* directions do not match the inferred local geomagnetic field direction at any subsequent time. Moreover, the Ashgillian minor intrusions show a reversal of magnetization which is explicable in terms of intrusion sequence and field reversal; anomalous directions can be explained as transitional. The polarity and direction of remanence in the intrusions is therefore most likely to date from the time of cooling. We do not know, however, whether it is preserved today as a genuine Thermal Remanent Magnetization (TRM) or as a diagenetic Chemical Remanent Magnetization (CRM) maintaining the primary remanence direction. In the Builth Volcanic Series the latter alternative is considered more probable because the intensities are comparable in the lavas and tuffs.

In addition we note that the lower pyroclastic component of the Series experienced extensive soda metasomatism during formation and is petrographically

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similar to the lavas. It is likely that this autometasomatism remagnetized the lower part of the sequence and may be the reason why the Series is of one polarity only.

The result presented here for the Builth Volcanic Series should be taken to supersede that of Nesbitt (1967) because it is based on a much larger collection and more exhaustive stability tests. This pole, dated as late Llanvirn, and that from the early Ashgillian intrusives are significantly different at the 95 per cent confidence level using the test of Watson (1956). They are discussed in the light of other British Ordovician data in Paper VI.

Department of Earth Sciences,
Leeds University,
Leeds, LS2 9JT

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